**1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**

**ANS:-**

Wireshark Packet Identification:

After refreshing the Flipkart homepage in Wireshark, I identified one packet for each layer by checking the Protocol and header information.

| Layer | Packet / Protocol | Header Details | Purpose |
|---|---|---|---|
| Application | HTTP/HTTPS | HTTP request / TLS information | Handles communication between the browser and Flipkart website. |
| Transport | TCP | Source Port, Destination Port, Sequence Number | Provides reliable delivery of data between devices. |
| Internet | IP | Source IP, Destination IP, TTL | Provides addressing and routes packets across networks. |
| Network Access | Ethernet | Source MAC, Destination MAC, EtherType | Delivers frames between devices on the local network. |

**2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).**

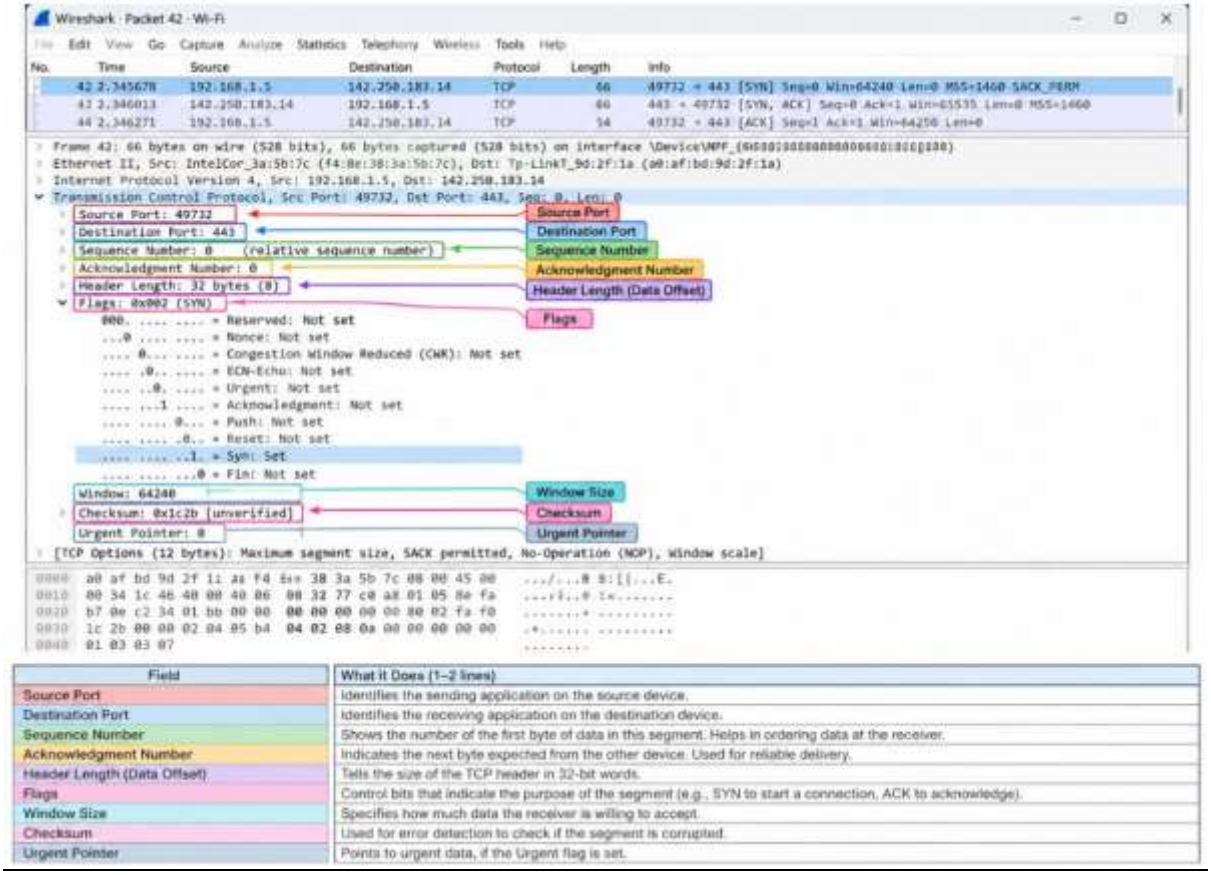
**ANS:-**

| OSI model layer | TCP/IP model layer | Real-world example |
|---|---|---|
| Application (7) | Application | WhatsApp messages |
| Presentation (6) | Application | Instagram image formatting and encryption |
| Session (5) | Application | Maintaining a Spotify login session |
| Transport (4) | Transport | TCP/UDP communication during a Zomato order |
| Network (3) | Internet | Google Maps routing using IP |

| | | |
|---|---|---|
| Data Link (2) | Network Access | Wi-Fi communication with a router |
| Physical (1) | Network Access | Wi-Fi radio signals or Ethernet cables |

**3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.**

**ANS:-**




| Field | What it Does (1–2 lines) |
|---|---|
| Source Port | Identifies the sending application on the source device. |
| Destination Port | Identifies the receiving application on the destination device. |
| Sequence Number | Shows the number of the first byte of data in this segment. Helps in ordering data at the receiver. |
| Acknowledgment Number | Indicates the next byte expected from the other device. Used for reliable delivery. |
| Header Length (Data Offset) | Tells the size of the TCP header in 32-bit words. |
| Flags | Control bits that indicate the purpose of the segment (e.g., SYN to start a connection, ACK to acknowledge). |
| Window Size | Specifies how much data the receiver is willing to accept. |
| Checksum | Used for error detection to check if the segment is corrupted. |
| Urgent Pointer | Points to urgent data, if the Urgent flag is set. |

***Figure : TCP Segment Header with Transport Layer Fields.***

| TCP Header Field | What It Does |
|---|---|
| **Source Port** | **Identifies the application sending the data.** |
| **Destination Port** | **Identifies the application that should receive the data.** |

| TCP Header Field | What It Does |
|---|---|
| **Sequence Number** | **Keeps track of the order of data so it can be correctly reassembled.** |
| **Acknowledgment Number** | **Indicates the next data expected from the other device.** |
| **Header Length** | **Shows the size of the TCP header.** |
| **Flags** | **Controls TCP communication, such as starting or acknowledging a connection.** |
| **Window Size** | **Shows how much data the receiver can accept at a time.** |
| **Checksum** | **Checks whether the TCP segment has been corrupted during transmission.** |
| **Urgent Pointer** | **Points to urgent data when the URG flag is used.** |

**4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).**

**ANS:-**

Flow shown in the diagram:

Application Layer: Instagram message is created.

Transport Layer: TCP breaks the message into segments and adds port numbers.

Internet Layer: IP addresses are added and packets are formed.

Network Access Layer: Data is sent as frames over Wi-Fi/mobile network.

Instagram Server: The data is received, reassembled, and the message is processed.

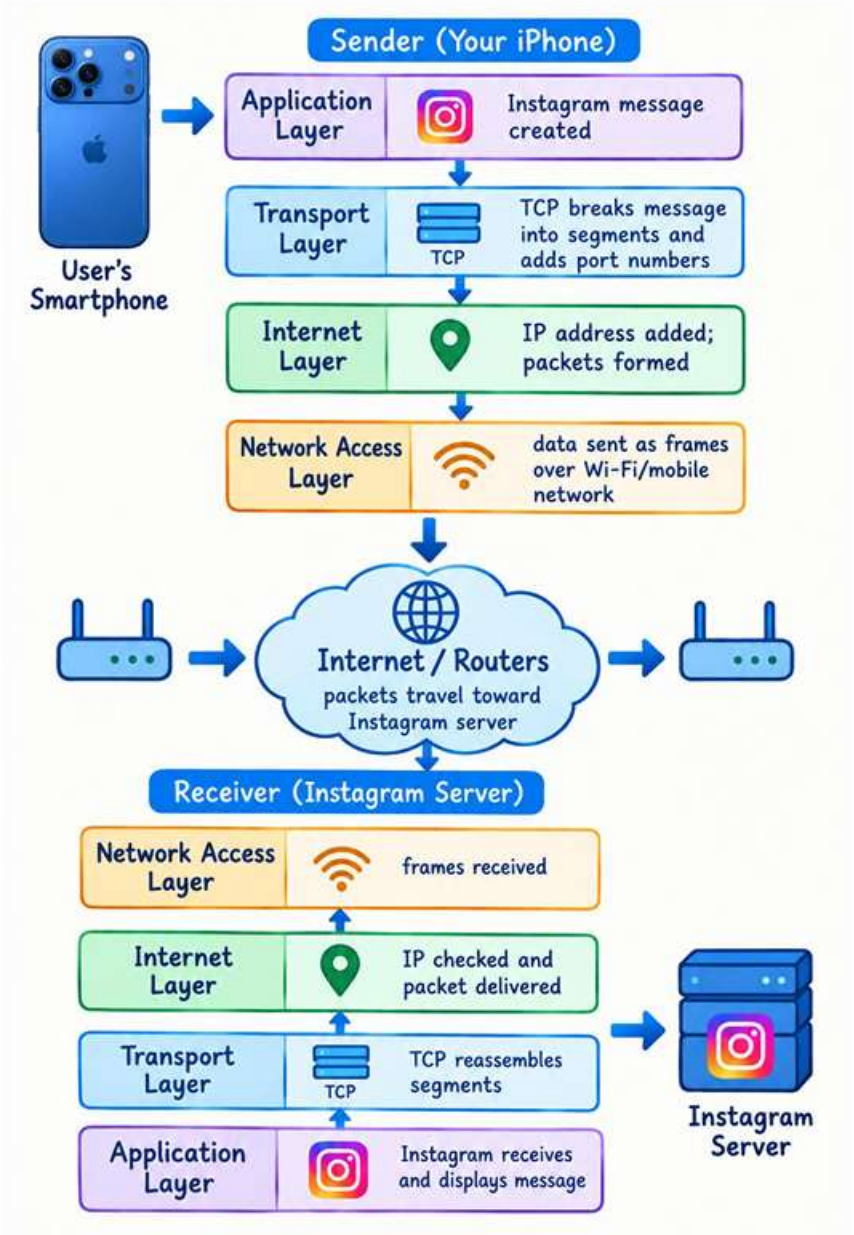



*Figure : How an Instagram Message Travels Through the TCP/IP Layers.*

**5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.**

**ANS:-**

**Wireshark Packet Analysis:**

**Selected Packet: TCP**

**TCP/IP Layer: Transport Layer**

**✍ How I Identified It:**

**I identified the packet as TCP by checking the Protocol column in Wireshark. The packet also showed Source Port and Destination Port, which are TCP header fields.**

**א Real-World Action:**

**This packet represents the reliable transfer of data when sending a message on WhatsApp. TCP helps deliver the message data correctly between the device and the server.**